

Rohith Pallamreddy

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science, GPA: 4.0

Atlanta, GA

May 2027

EXPERIENCE

Incoming at Cruise (General Motors Autonomous Vehicles)

Warren, MI

Autonomous Vehicle Software Engineer Intern (Platform OS and CPU team)

Summer 2026

- Will develop and **optimize embedded software platform** components including drivers, operating system, BSP, and software stack for high-performance SoCs powering ADAS and **autonomous driving applications with C++ and RTOS principles**.
- Will work on real-time embedded platform solutions to improve performance, reliability, and **scalability of the core software stack** for production autonomous vehicle systems.

Resilient and Equitable Mobility Analytics and Planning @ Georgia Tech

Atlanta, GA

Machine Learning Researcher (Mapping and Mobility)

Jan 2025 – Present

- Generated spatiotemporal processing pipelines to efficiently compress and extract **70+ GB** of data into **<1 GB** for high-performance mobility analysis
- Architected **algorithmic trajectory reconstruction** pipelines using Kalman filters and ML, achieving **96.3% accuracy**, providing applications for high-fidelity **Autonomous Vehicle path planning** and localization
- Generated predictive Motif pipelines with **graph-based models** to identify human decision-making features with **84% accuracy**, providing applications for **human-like routing in Autonomous Vehicles** to increase passenger comfort and the perception of a **safe, human-driven experience**

Perception and Simulation Engineer (Microclimate)

Aug 2025 - May 2026

- Processed **500+ GB of LiDAR datasets** to recreate 3-dimensional terrain, building, and canopy information for downstream environmental simulation
- Utilized LiDAR derived features to implement SOLWEIG to **simulate and predict** perceived microclimate (RMSE < 1.5°C), enabling **thermal-aware routing** to mitigate **Autonomous Vehicle hardware overheating**
- Engineered and debugged alternative SOLWEIG pipeline to simulate and predict perceived temperature with a **93% faster runtime**, no spatial compression, and **99.3% accuracy**

PROJECTS

RuncHypervisor (Realtime Unmanned Navigation and Control Hypervisor)

May 2026 – Present

- Architected a **bare-metal Type-1 ARMv8-A** hypervisor **BSP**, leveraging **hardware virtualization extensions** and custom **3-level Stage-2 page tables** to enforce memory isolation across a **112 MB** guest space.
- Designed a **deterministic mixed-criticality RTOS scheduler** for **16ms/64ms** tasks using low-level **hardware timers**, optimizing **context-switch latency** by instantly swapping **34** registers during device **MMIO faults**.

RuncOS (Realtime Unmanned Navigation and Control Operating System)

Dec 2025 – May 2026

- Developed a **bare-metal Autonomous Vehicle C++ RTOS** on ARM Cortex-M3; featuring a **1,000 Hz SysTick** scheduler that **achieves <5 μ s context-switch latency** across all concurrent threads.
- Architected a **sub-1ms** sensor-to-actuator control loop using **Q16.16 fixed-point math** for autonomous steering and braking; integrated **safety-critical monitors** for fault detection and state-restoration recovery

Motorcycle HIL Simulator

Aug 2025 – Dec 2025

- Developed a Hardware-in-the-Loop (**HIL**) system integrating Arduino-based IMU sensors with a **custom UE4 physics model** to achieve **98%** fidelity in real-time lean angle replication.
- Implemented a **Kalman Filter** to fuse noisy accelerometer and gyroscope data, reducing signal variance by **62%** and ensuring stable < 10ms latency for high-speed simulation dynamics.

TECHNICAL SKILLS

Languages: C++20 (STL, Boost, Templates, RAII, GDB), Python (C-Extensions, Cython), C, Rust, SQL, Bash
Systems & Sim: High-Fidelity Physics (PyBullet, UE4), HIL/SIL Testing, ROS2 (Pub/Sub, ActionLib), Gazebo
Perception & Math: Computer Vision (OpenCV, MediaPipe), State Estimation (Kalman/EKF, SLAM), Eigen, PCL
ML & Optimization: PyTorch (TensorRT, CUDA Kernels), Scikit-Learn, Distributed Training, Latency Optimization
Tooling & Infra: Linux Kernel/Systems, Bazel (Hermetic Builds), Docker, CI/CD, Git, GCP, Valgrind, Perf